



Increasing Usability of Ankle Exoskeletons Through a Shoe-binding Design

Joelle Yoon, Jae-Ryeong Choi, Patrick Slade, Harvard SEAS

Introduction

Background

- Ankle exoskeletons are proven to effectively increase mobility through lowering metabolic costs for the user.
- Many designs rely on modifying the user's shoe to anchor the ankle exoskeleton, lowering usability¹.

Objective

The purpose of this project is to design a non-invasive ankle exoskeleton interface that can easily be attached to the user's shoe without requiring any shoe modifications.

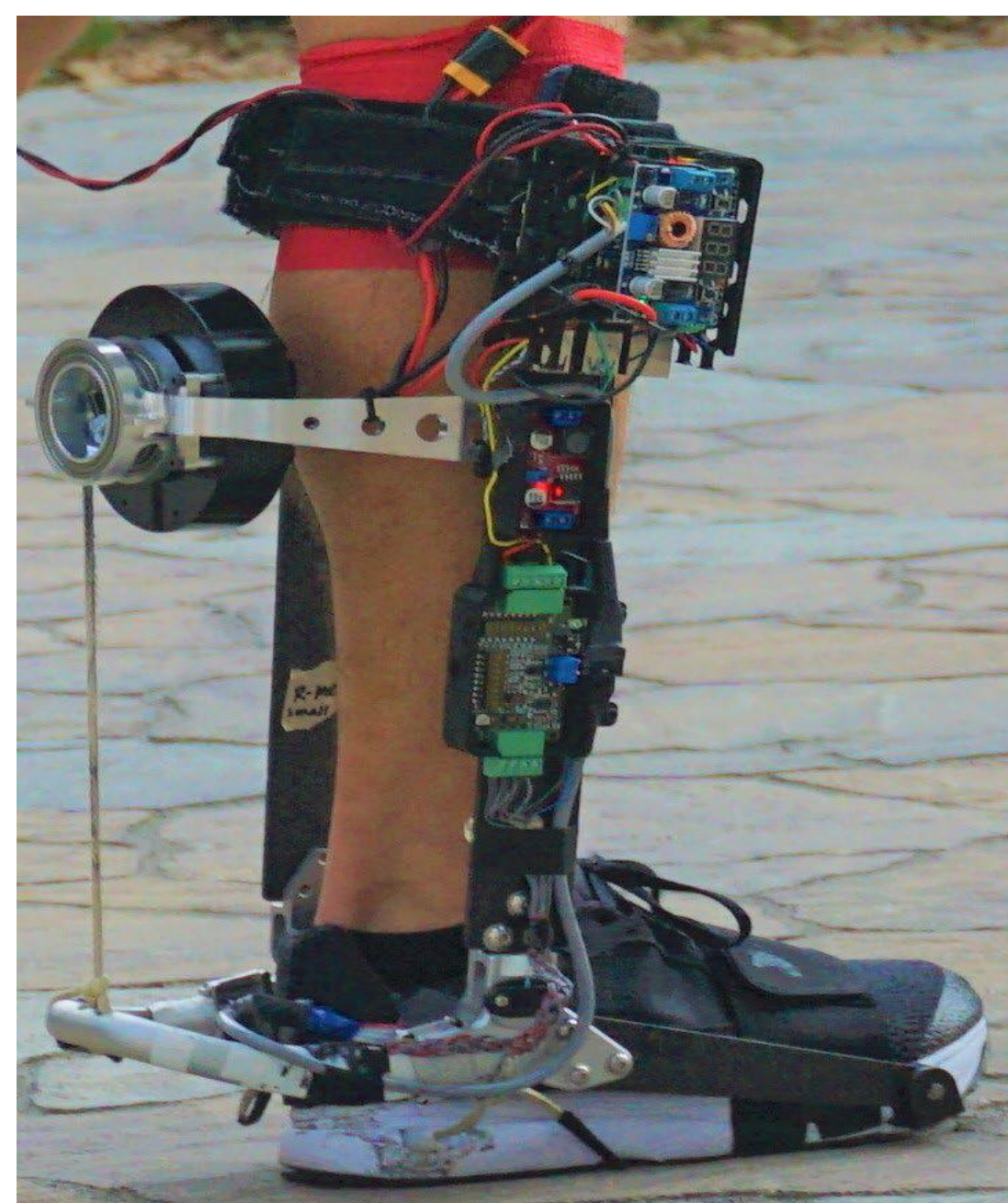


Figure 1. Rigid, powered ankle exoskeleton made at Stanford with anchoring points inserted into the user's shoe².



Figure 2. Rigid, active ankle exoskeleton developed by MIT also requiring shoe modifications at the anchoring points³.

Acknowledgements

This work was supported by the Harvard College Office of Undergraduate Research and Fellowships. We thank Andrew Chin for help with data collection and analysis, as well as everyone at the Slade Lab for their support.

Methods

Design

- **Non-invasive anchoring system**
 - Inspiration: plantar fascia
 - Side-adjustable, shoe-binding design
 - ~140mm moment arm
- **Heel piece:** designed and fabricated using CAD and 3D printing software



Figure 3. Rear view from the side.



Figure 4. Side view.



Figure 5. Bottom view.

Device Characterization:

- **Purpose:** Torque transmission
- **Data collection:** Motion capture system (Qualisys, USA) with markers
- **Force:** load cell (Futek, USA)
- Results to be compared to fully rigid ankle exoskeleton

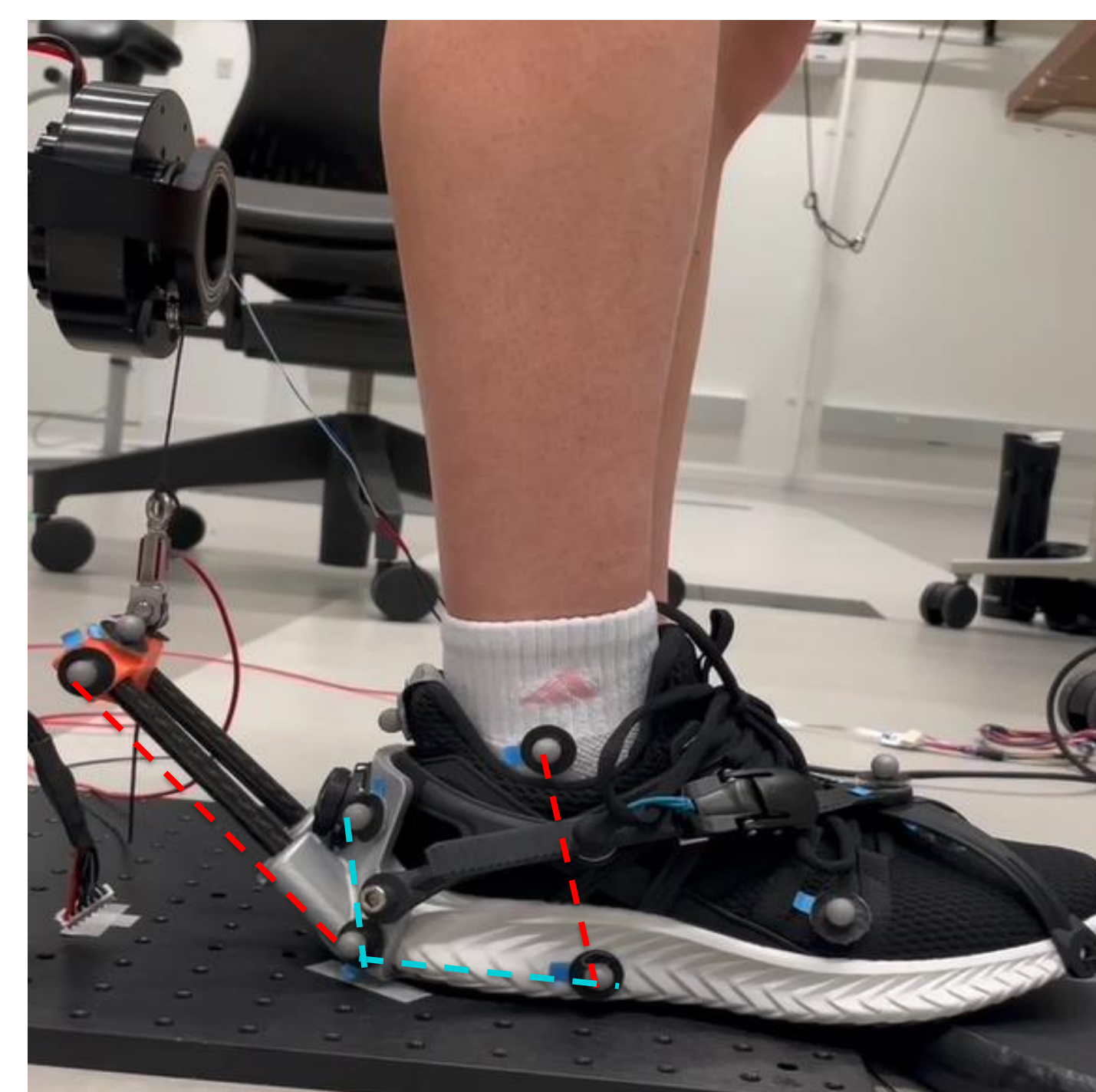


Figure 6. Side view of experimental set up (no force applied).



Figure 7. Motion capture data (XZ-plane). Blue dashed lines = radial deformation of rigid heel structure; red dashed lines = radial deformation of moment arm.

Results

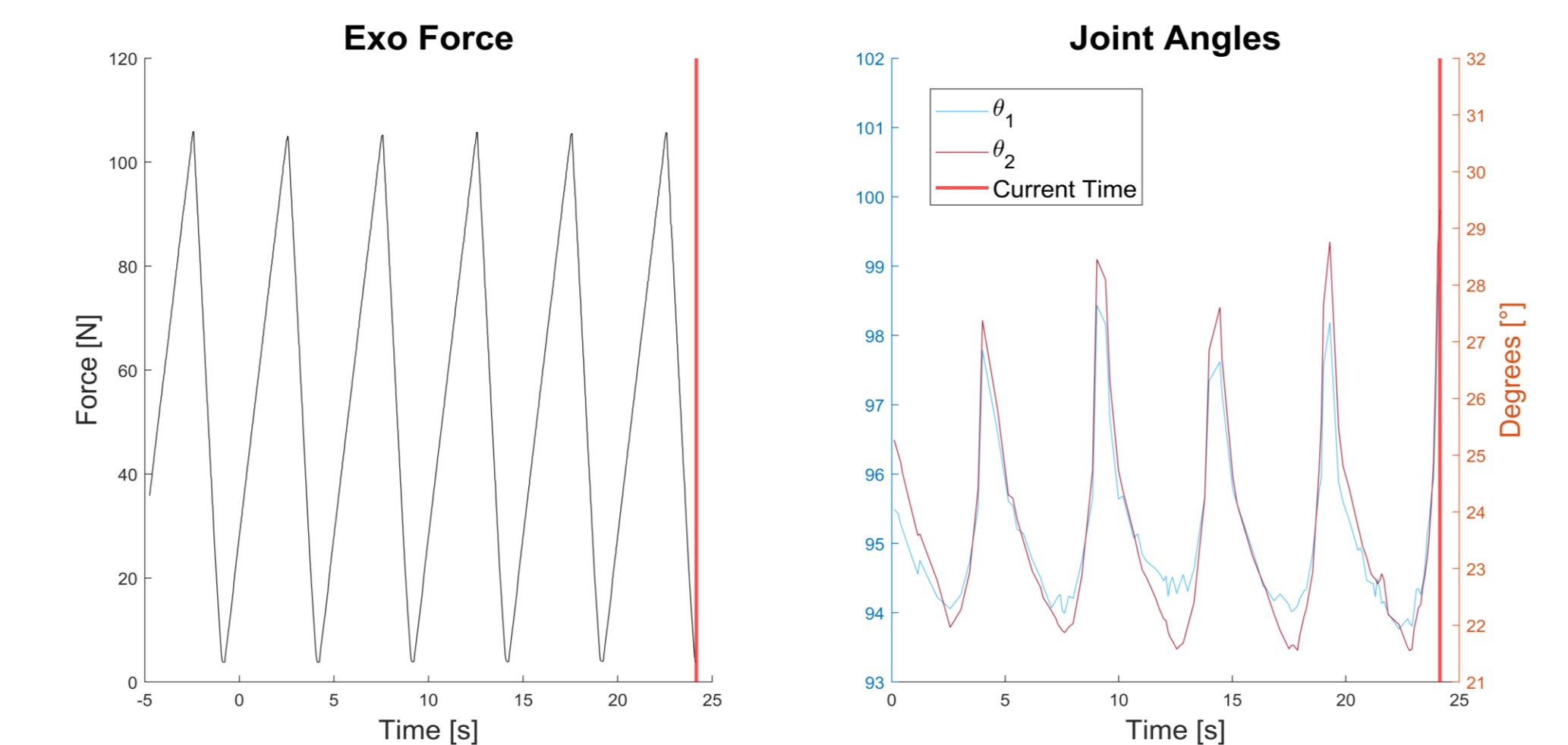


Figure 8. Left: force vs. time; right: theta vs. time (min force with tightened configuration).

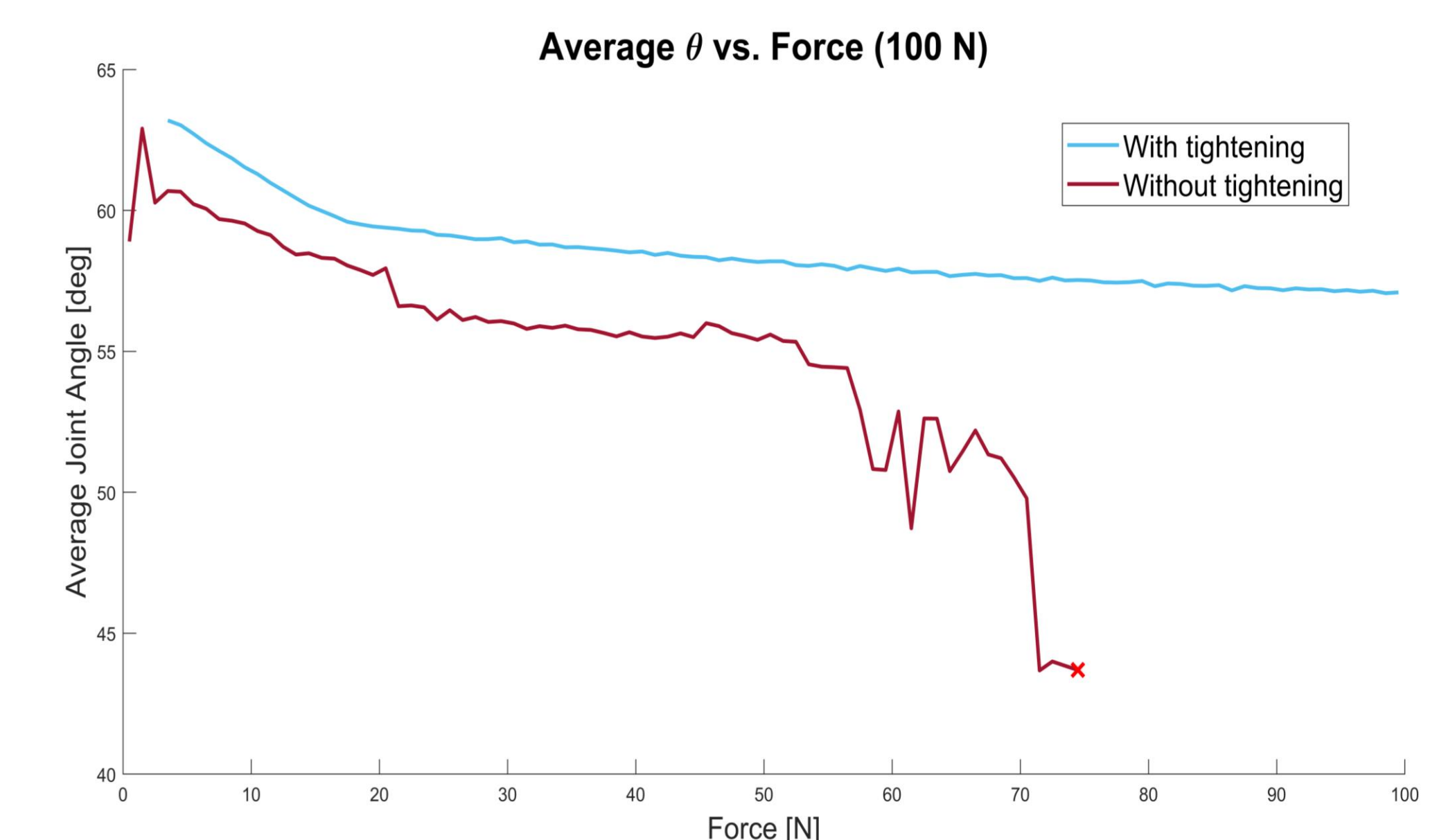


Figure 9. Average joint angle vs. force (100 N) for two trials. Blue = tightened configuration; red = untightened configuration.

- Initial data analysis indicates the need for shoe-binding components in delivering torque assistance.

Conclusion

- Preliminary results suggest that the exoskeleton is able to provide sufficient torque assistance during walking.

Next Steps

- Collect data on torque transmission and shoe deformation to verify design, making design modifications as needed
- Design and implement full exoskeleton set up for running human experiments

References

1. Yandell, M., Tacca, J., & Zelik, K. Design of a Low Profile, Unpowered Ankle Exoskeleton That Fits Under Clothes: Overcoming Practical Barriers to Widespread Societal Adoption. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 27, 712-723 (2019).
2. Slade, P., Kochenderfer, M.J., Delp, S.L. et al. Personalizing exoskeleton assistance while walking in the real world. *Nature* 610, 277-282 (2022).
3. Mooney, L.M., Rouse, E.J. & Herr, H.M. Autonomous exoskeleton reduces metabolic cost of human walking. *Journal of NeuroEngineering and Rehabilitation* 11, 151 (2014).

Contact Information/Learn More

Joelle Yoon
jdyoon@college.harvard.edu

